```
  "write": {
    "type": "fixed",
    "size": 4,
    "queue_size": 10000
  },
..
}
```

As you can see, Elasticsearch assigned seven and then four threads for searching and writing operations automatically, based on my computer's available number of processors. These settings are static, so we must edit the configuration file and restart the server for them to take effect.

In addition to the previous recommendations, there are a few others, such as increasing the indexing buffer size, switching off replicas during indexing, disabling swapping, and so forth. You can find a list of recommendations for improving the performance of indexing operations at http://mng.bz/a19Y.

15.3 Unstable clusters

A common problem when working with Elasticsearch is the stability of the cluster. Keeping the cluster healthy helps when servicing clients' requests. Many things can go wrong with the cluster! In this section, we look at some of the most common concerns.

15.3.1 Cluster is not GREEN

A cluster's health is represented by an effective, high-level traffic light status. Issuing `GET _cluster/health` provides the real-time status of the cluster: RED, YELLOW, or GREEN. An administrator's primary job is to make the cluster's status always GREEN.

If the status is RED, it must be addressed at any cost because part of the cluster is either not functioning or down. This unhealthy state of the cluster (indicated by the RED status) may be due to any number of problems such as hardware failures, network outages, corrupted filesystem, and others, leading to a loss of cluster nodes. In such a situation, DevOps engineers stop everything they are doing to fix this problem.

A YELLOW status indicates that the cluster is unhealthy but can run with manageable risk. This can indicate a loss of a few nodes, unassigned shards and replicas, or other problems. Although the cluster is available for business, it may soon go into the RED state if we don't fix the problem. The YELLOW state means trouble is waiting to happen.

Regular heartbeats originating from masters to nodes and vice versa establish the overall health of the cluster; the heartbeats from a master node to other nodes in the cluster provide a mechanism to find disconnected or nonresponsive nodes. In the same way, all nodes ping the master node regularly to check if it is alive and running.

The master node takes appropriate actions such as cluster data redistribution, cross-cluster replications, index management, shard rearrangement, and so on when members of the cluster appear or disappear. When they are deemed unfit for the purpose at hand, node(s) are removed from the cluster by the master.